

B.A./B.Sc. (Hons.) Semester IV Examination 2022-23
STATISTICS

Paper No. STA-402: Practical based on Course No. STA-401

Time: Three hours

Max. Marks: 70

NOTE: Attempt any three questions. All questions carry equal marks. Show all layout designs and computational tables clearly.

1. Assume that a population consists of 7 similar shipping containers having the following weights (in kilograms):

9.6, 10.2, 10.4, 9.8, 10.0, 10.2, 9.6

- (a) Find the exact population mean μ and the population standard deviation σ .
- (b) Draw all possible random samples of size $n = 2$ containers: (i) without replacement (SRSWOR), and (ii) with replacement (SRSWR). Calculate the sample mean weight $\bar{x}$ for each drawn sample.
- (c) Construct the empirical sampling distribution of $\bar{x}$ under both schemes.
- (d) Calculate the mean and standard error of the sampling distribution directly and verify against theoretical formulas.

2. (a) Suppose a finite population of size $N = 9$ is stratified into 3 distinct strata with the following subpopulation measurements:

Stratum I: $X_{11} = 1, X_{12} = 2, X_{13} = 4$

Stratum II: $X_{21} = 6, X_{22} = 8$

Stratum III: $X_{31} = 11, X_{32} = 15, X_{33} = 16, X_{34} = 19$

If a sample of size $n = 6$ is chosen by selecting exactly two units from each stratum, calculate how many unique stratified samples can be chosen. List these samples and compute the stratified sample mean for each.

- (b) Select a stratified random sample of size $n = 5$ by proportional allocation from the following population, and provide the estimate of the population mean:

Stratum I: 12, 14, 13, 8

Stratum II: 25, 30, 40, 35, 24, 28

3. The final marks received by students in three sections of an elementary statistics course meeting at different times of the day are given below:

8-o'clock: 121, 117, 145, 108, 142, 154, 115, 81, 122, 127, 122

10-o'clock: 97, 145, 119, 189, 143, 133, 149, 107, 154

2-o'clock: 134, 89, 108, 88, 146, 153, 130, 144, 125, 111, 87, 162

Is there a statistically significant difference in the scores received by the students across different times of the day? State completely your null and alternative hypotheses, ANOVA tables, and final conclusions.

4. Five agricultural fertilizers (A, B, C, D, and E) were evaluated by arranging plots in a 5×5 Latin Square Design (LSD) in an open field. The operational row/column layout along with final yield values (in bushels per plot) is given below:

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	B (4.9)	D (6.4)	E (3.3)	A (9.5)	C (11.8)
Row 2	C (9.3)	A (4.0)	B (6.2)	E (5.1)	D (5.4)
Row 3	D (7.6)	C (15.4)	A (4.0)	B (6.0)	E (4.6)
Row 4	E (5.3)	B (7.6)	C (13.2)	D (8.6)	A (4.9)
Row 5	A (9.3)	E (6.3)	D (11.8)	C (15.9)	B (7.6)

Analyze the full dataset for concrete evidence at a 5% level of significance that the mean yields are not equal for the five fertilizers, and determine the single best variety.

5. Using the United States agricultural bank database provided in the appendix (comprising 50 items matching Non-real estate farm loans (X) and Real estate farm loans (Y)), select a representative SRSWOR sample of eighteen states.
 - (a) Estimate the average real estate farm loans ($\bar{Y}$) using the ratio method of estimation, given that the population true average amount of non-real estate farm loans ($\bar{X}$) is known to be \$879.16.
 - (b) Estimate the same parameter using the linear regression method of estimation.
 - (c) Find the approximate mean square error (MSE) of the ratio estimator and comment on its precision.
 - (d) Find the approximate mean square error (MSE) of the regression estimator.

Appendix: Population Database for Question 5

No.	State	X (Non-Real)	Y (Real)	No.	State	X (Non-Real)	Y (Real)
1	AL	348.33	408.98	26	MT	722.03	292.97
2	AK	3.43	2.61	27	NE	3585.41	1337.85
3	AZ	431.44	54.63	28	NV	16.71	5.86
4	AR	848.32	907.70	29	NH	0.47	6.04
5	CA	3928.73	1343.46	30	NJ	27.51	39.86
6	CO	906.28	315.81	31	NM	274.04	140.58
7	CT	4.37	7.13	32	NY	426.27	201.63
8	DE	43.23	42.81	33	NC	494.73	639.57
9	FL	464.52	825.75	34	ND	1241.37	449.10
10	GA	540.70	939.46	35	OH	635.77	870.72
11	HI	38.07	40.78	36	OK	1716.09	612.11
12	ID	1006.04	53.75	37	OR	571.49	114.90
13	IL	2610.57	2131.05	38	PA	298.35	756.17
14	IN	1022.78	1213.02	39	RI	0.23	1.61
15	IA	3909.74	2327.03	40	SC	80.75	87.95
16	KS	2580.30	1049.83	41	SD	1692.82	413.78
17	KY	557.66	1045.11	42	TN	388.87	553.27
18	LA	405.80	282.57	43	TX	3520.36	1248.76
19	ME	51.54	8.85	44	UT	197.24	56.91
20	MD	57.68	139.63	45	VT	19.36	57.75
21	MA	56.47	7.59	46	VA	188.48	321.58
22	MI	440.52	323.03	47	WA	1228.61	1100.75
23	MN	2466.89	1354.77	48	WV	29.29	99.28
24	MS	549.55	627.01	49	WI	1372.44	1229.75
25	MO	1519.99	1579.69	50	WY	386.48	100.96

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>